



Department of Electrical and Computer Engineering
COMSATS University Islamabad, Attock Campus

Lab Rubrics Form	
Lab Name	Amplitude Shift keying
Lab Number	10
Submitted by:	
Student Names	Registration #
Sumaira Safeer	FA22-BCE-019

Rubrics		Marks		
		1	2	3
PLO-5 Modern Tools Usage	Ability to execute the experiment utilizing hardware, software tools, or a combination of both.			
PLO-10 Communication Skills	Present lab activities effectively through detailed lab reports, oral examinations, and engaging presentations.			

LAB: 10

Topic: Amplitude Shift keying

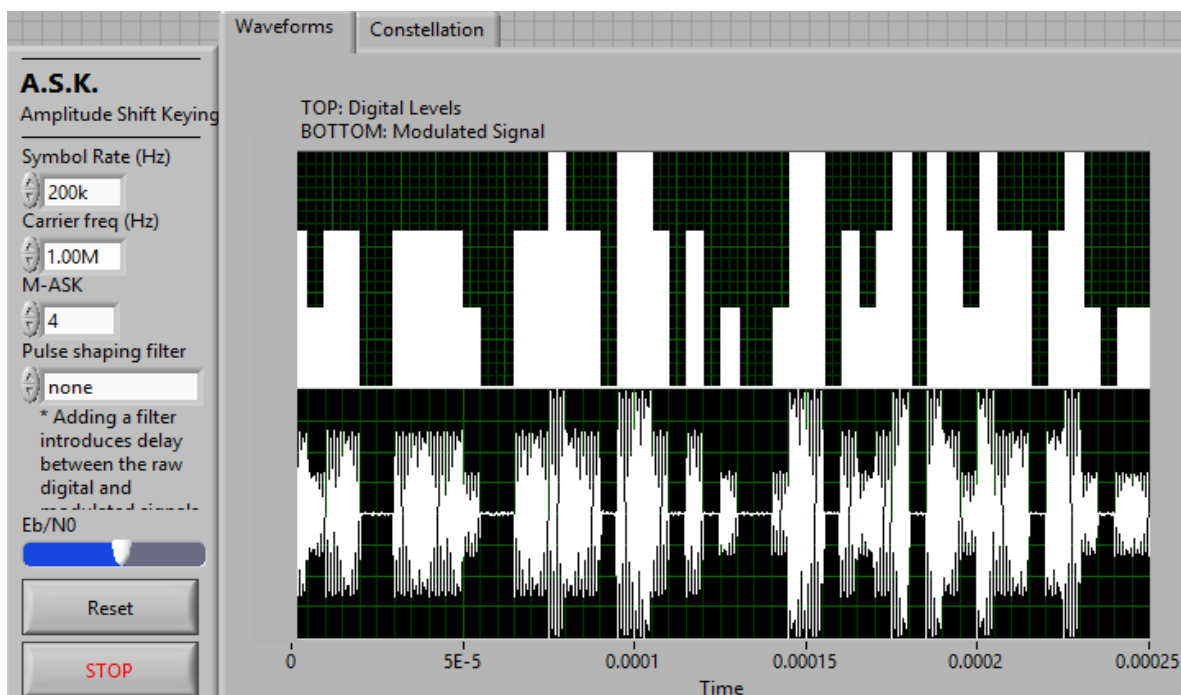
Objective :

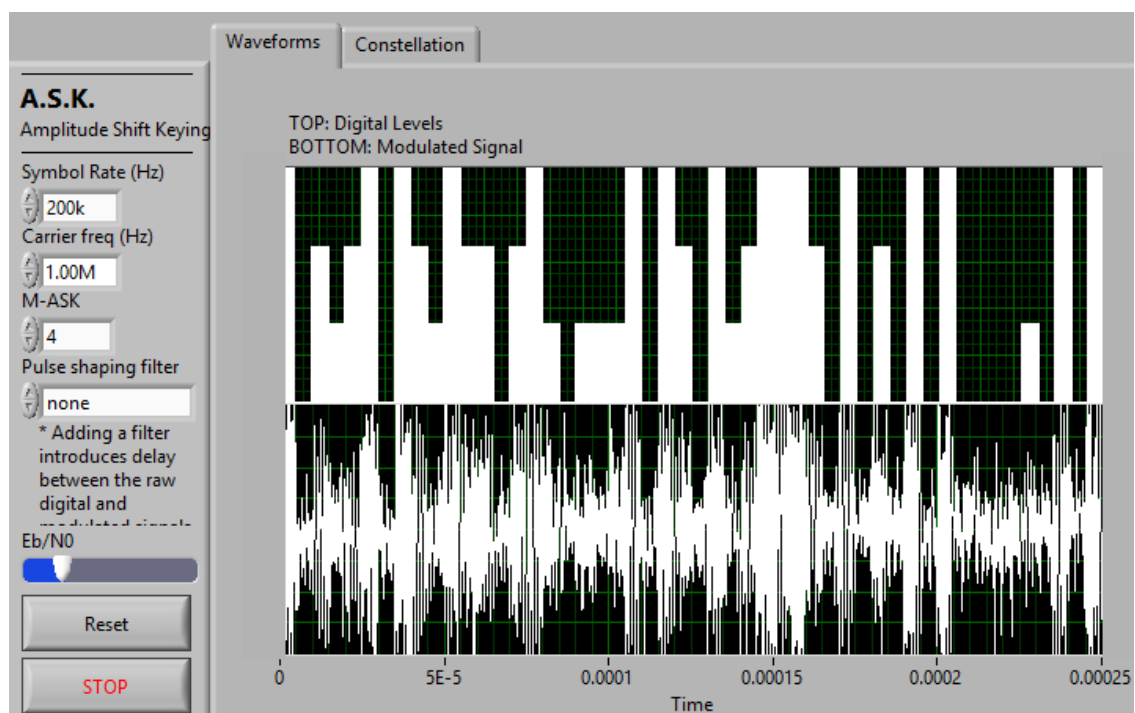
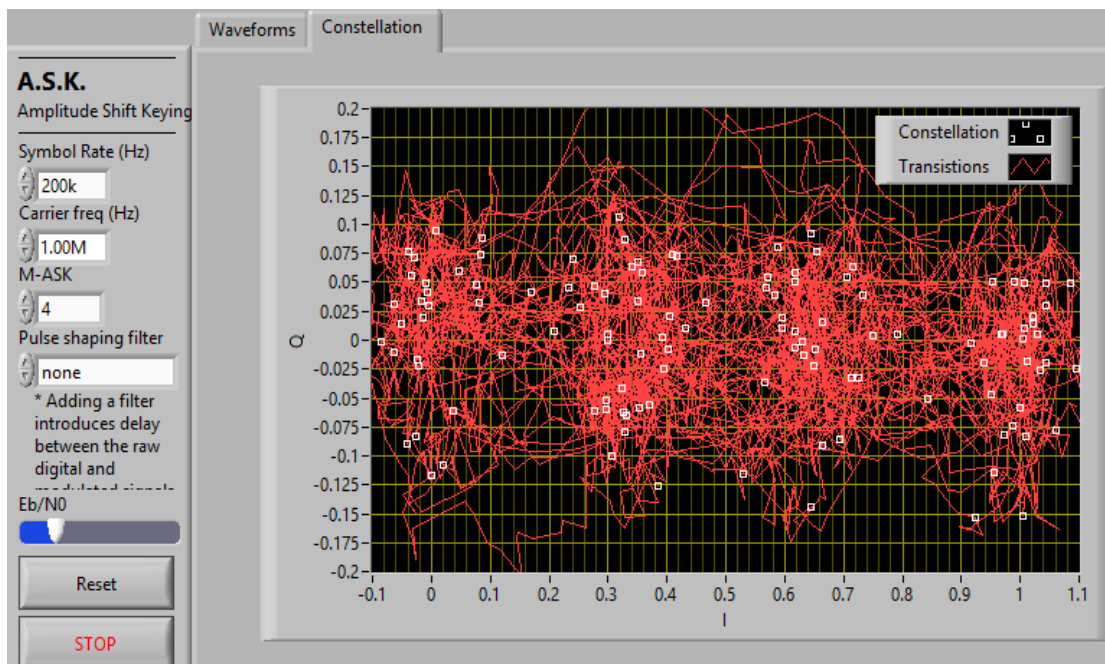
Implementation of amplitude shift keying in LabVIEW.

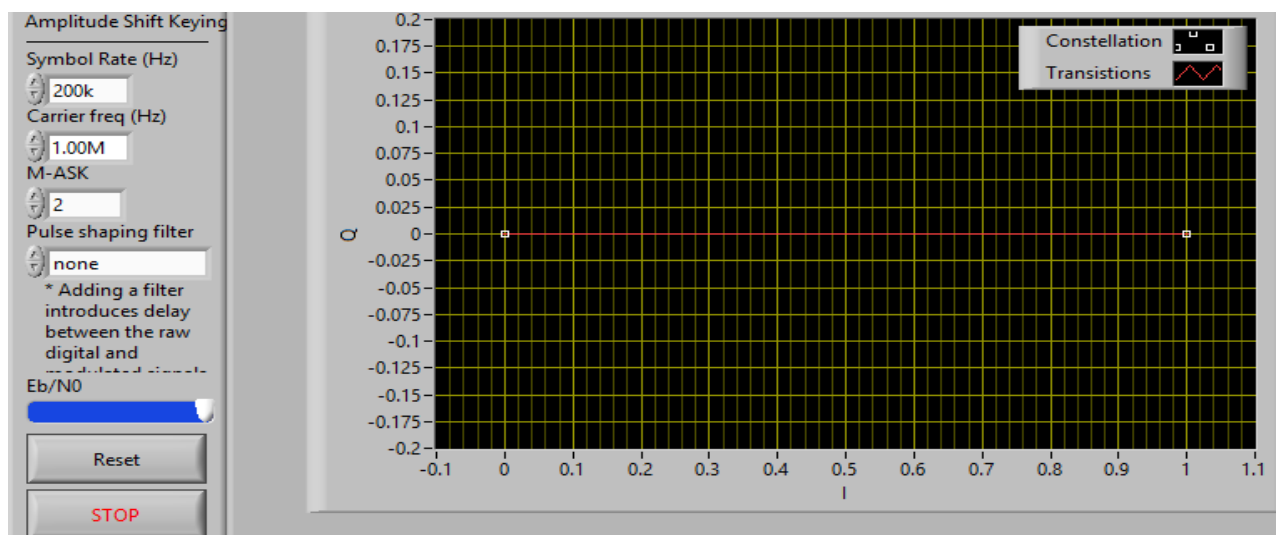
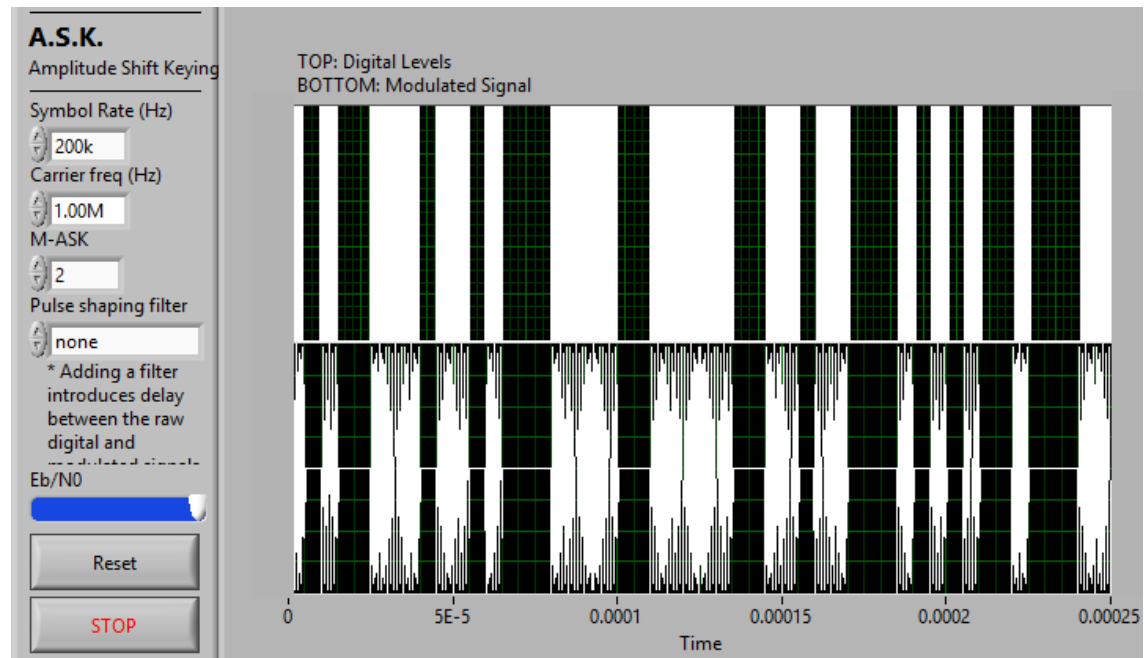
Introduction:

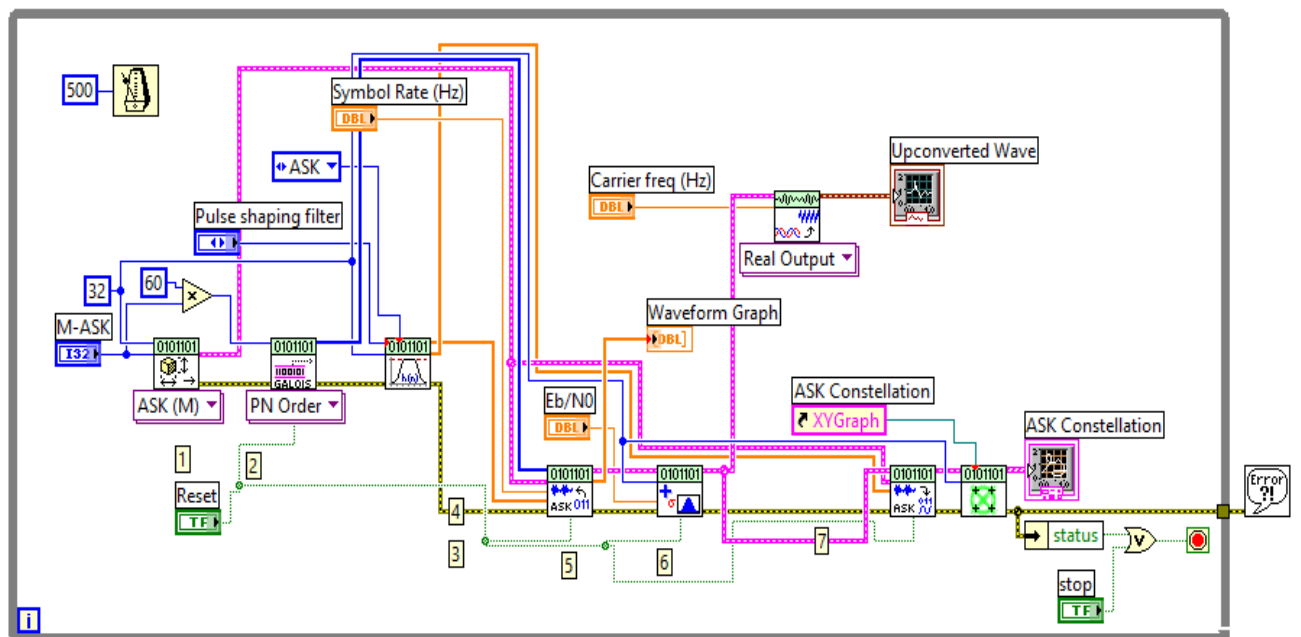
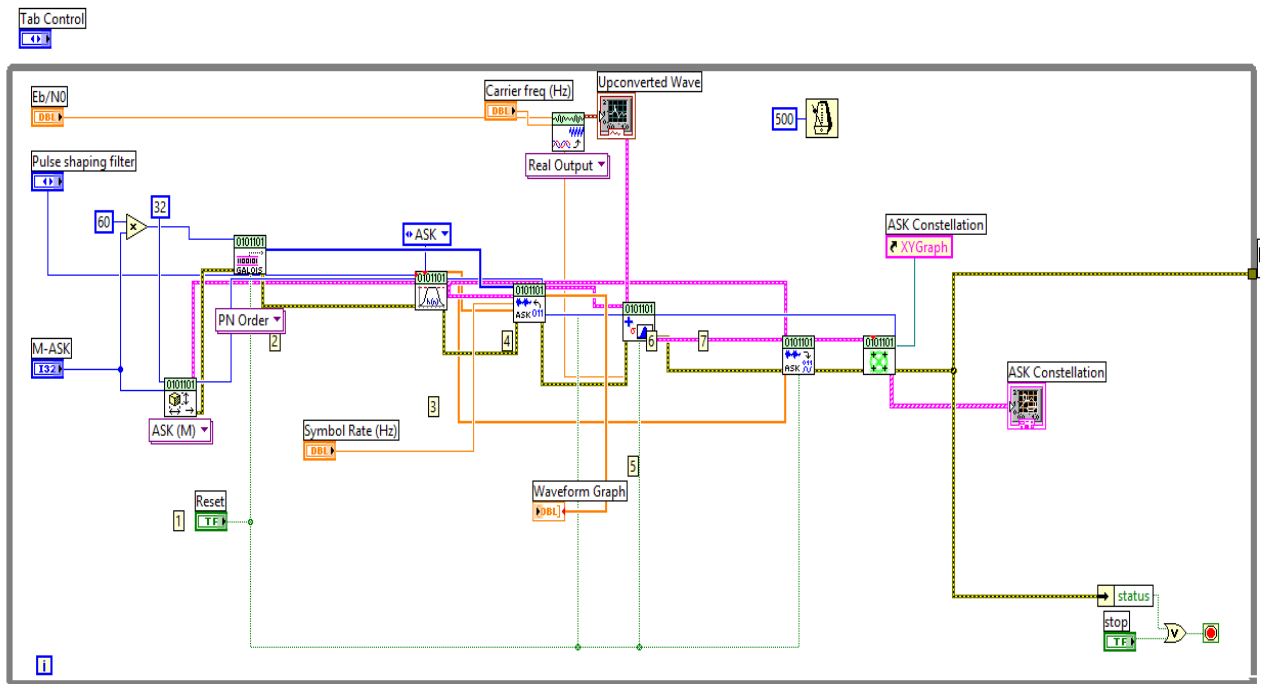
Fundamentally, digital modulation requires changing characteristics of the carrier wave over time. Each change results in a sine wave with a different phase, amplitude, or frequency than before. As a result, different “states” of the sine wave are referred to as symbols which represent some digital bit pattern. In this exercise, we will construct a LabVIEW VI that transmits and receives a digital bit stream in software using ASK.

FRONT PANEL:









Conclusion:

This lab focused on implementing Amplitude Shift Keying (ASK) in LabVIEW, a digital modulation technique where the amplitude of a carrier signal varies based on the digital bit stream. By constructing a LabVIEW Virtual Instrument (VI), we transmitted and received a digital bit stream using ASK. The exercise provided a hands-on understanding of how amplitude variations represent binary data, and how ASK modulation and demodulation work. Overall, this lab demonstrated how digital information can be encoded and decoded using changes in the amplitude of a carrier wave.